DS302 Lab 2

February 5, 2025

1. Look at the `HelloWorld.asm` program. Understand how the string constant is stored in memory.
2. See the code `simple.asm` and understand how various data types are represented.
3. See the `loop.asm` program. Look up the instructions `bne, j`.
4. Load and execute the `inpout.asm` program. Look up the documentation of the various system calls used in the program.
5. Write a program to add the numbers in an array and store the sum in a memory location. Use the `.word` directive. See page A-49. See also `sw` instruction.
6. Write a program to find the minimum number in an array and store it in a memory location. You can use `bnez, bge`.
7. Write a program that uses a procedure call to compute x^y, where x and y are input integers. Create a procedure that accepts x and y as arguments and returns the result. Within the procedure, you can use a loop. See the skeleton code `proc_skel.asm`, and also the example in the lecture slides on Moodle.